

our purpose. Fixed voltages of 5 V and 12 V are obtained from 7805 and 7812 ICs, respectively.

LM338, 7805 and 7812 should be mounted on proper heat-sinks. However, heat-sink of LM338 should not

touch the heat-sink of the other two regulators. This is because the metal cap (body) of LM338 is connected to its pin 2 (V_{out}) internally, while the metal cap of 7805 and 7812 is connected to the ground pin. If 7805 or 7812 touches LM338, it will lead to short circuit of LM338.

When external capacitors are used, it is sometimes necessary to add protection diodes to prevent the capacitors from discharging through low-current points into the regulator. Most 20 μ F capacitors have low enough internal series resistance to deliver 20A spikes when shorted. Although the surge is short, there is enough energy to damage parts of the IC.

In operation, LM338 develops a nominal reference voltage of 1.25 V, V_{REF} between the output and adjustment terminal. The reference voltage is impressed across program resistor R1, which is of 240 ohms, and an external potentiometer of 5 kilo-ohms is connected between the adjust pin and ground.

Interface and display circuit description

Fig. 5 shows the interface and display circuit. This stage consists of a step-down transformer (X2), AVR microcontroller ATmega8, serial interface MAX232, op-amp 741, 16 $\times$ 2 character LCD and a few other components.

A 230V AC to 9V, 500mA AC step-down transformer is sufficient to power this module. Using bridge rectifier module BR1, filter capacitors C7 and C8, and another 7805 regulator, we obtain a regulated 5V output for powering the microcontroller, LCD and MAX232.

Switches S2 and S3 are connected to port pins PC4 and PC5 of the ATmega8,

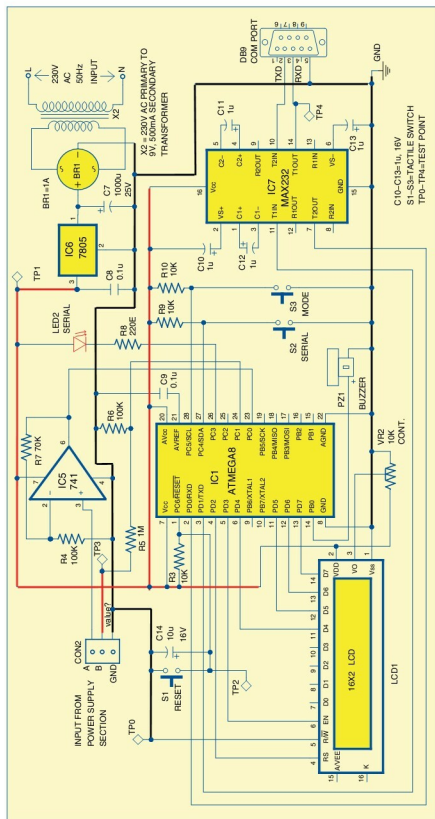


Fig. 5: The interface and display circuit